

Genetic sharing across sleep constructs reproduced estimates and interpretive limits

Investigator review note — outside the manuscript body. This is one integrated, qualified research draft, prepared against the evidence snapshot of 10 October 2026. It targets the format of a Nature Communications Article but does not establish the journal's required scientific advance or submission readiness. Independent automated methodological, mathematical and biological reviews found no admitted new method or new primary discovery. The draft describes reproduced historical associations, diagnostic sensitivities and limitations of source transport. Preprocessing comparisons for all 13 historical validation sources have completed with independently reviewed ordered-content/QC agreement. Separate core comparison attempts completed 5 of 27 and 7 of 35 source chains, respectively; the extension attempt completed 79 of 100. All three attempts ended in preserved partial states, with no worker active. These separately defined plans leave verification of the full source chains incomplete. The manuscript claim ledger and figure mapping are separate companion files. Bracketed administrative entries below require accountable human completion and are not research assertions.

Abstract

Sleep GWAS measure different complaints, behaviours and clinical outcomes, complicating interpretation of shared genetic architecture. We reconstructed a historical atlas of 12 sleep traits against 33 core and 100 extension phenotypes, retaining its original analysis families. Across 190 native jobs, 158 heritability and 1,637 genetic-correlation estimates reproduced available historical printed results. The 396-test core family contained 153 primary and eight sensitivity-tier false-discovery-rate positives; the 1,200-test extension contained 603 positives. Insomnia showed sharing with depression, frailty, adiposity and several gastrointestinal phenotypes, while inflammatory bowel disease had an opposing direction. Questionnaire and device duration estimates were descriptively different, without a calibrated difference test. Among 217 historical external candidates, 23 qualified outcome-side positives and 18 directional estimates reused the original sleep GWAS; 17 candidates were QC-ineligible and 159 unavailable. None established fully independent two-trait replication. Sixty-two sensitivity estimates exposed regression and sample-size dependencies without creating new discoveries. Clinical-insomnia heritability passed a necessary signal criterion under assumed effective sample size. Core and extension raw-source preprocessing remain incomplete. The resource supports source-defined sharing while delineating unresolved construct, ascertainment, overlap and joint-uncertainty requirements.

Introduction

Genome-wide association studies (GWAS) have expanded the evidence for genetic contributions to sleep complaints, timing, duration and device-derived behaviours. These measurements do not define a single phenotype. Frequent trouble initiating or maintaining sleep is distinct from a coded clinical diagnosis, reported hours of sleep and movement-derived estimates within a sleep-period window. The public UK Biobank insomnia release, reported-duration study and accelerometer sleep study illustrate these differences in measurement and sampling [1–3]. Genetic correlation can describe common-variant sharing between their observed phenotypes and disease outcomes, but its interpretation depends on the source populations, phenotype definitions and statistical information supporting each estimate [4].

Multidimensional sleep architecture and its relationships with health have already been studied using multivariate sleep domains, composite sleep-health scores and broader disease screens [5–7]. These precedents constrain claims of first identification. A reproducible association table also leaves questions that are essential for scientific interpretation: whether a repeated result uses the same processed input, whether an external study replaces one or both source populations, whether clinical ascertainment is compatible, and whether apparent differences between correlations have valid joint uncertainty. Source reconstruction, association reproduction and independent replication therefore require distinct evidence.

Here we integrate a numerical reconstruction of the historical Sleep GWAS Atlas and its 100-phenotype extension with source qualification and completed sensitivity analyses. We retain the original 396- and 1,200-test families rather than selecting a favourable subset for revised significance thresholds. We examine illustrative observed patterns across mood, adiposity, gastrointestinal disease and lipids, alongside the complete tables. We then distinguish qualified external outcome-source evidence from independent replication and assess what the present measurements can support. The aim is a transparent account of source-defined genetic sharing and its reproducibility limits. This study does not report a new causal mechanism, validated latent sleep construct or new statistical method.

Results

Numerical reproduction with preserved test families

The core panel contained 12 sleep traits and 33 outcomes; the extension paired the same sleep traits with 100 additional phenotypes. The sleep panel comprised frequent insomnia complaints, continuous reported duration, short and long duration categories, morningness, daytime sleepiness, napping, habitual snoring, coded sleep apnoea, and device-derived sleep efficiency, duration and midpoint. Native reproduction completed 57 core, 112 extension and 21 validation jobs, yielding 158 standalone heritability estimates and 1,637 genetic correlations (Fig. 1). Captured estimates agreed with the available historical logs at their printed precision. Historical unrounded values were unavailable, so this result does not establish equality to an original full-precision numerical object.

The complete core family retained 396 tests. At Benjamini–Hochberg false discovery rate (FDR) below 0.05, 153 results were in the original primary tier and eight in the sensitivity tier. The extension retained all 1,200 tests and 603 FDR positives. These counts describe the historical statistical screens, whose outcomes and sleep traits are correlated; they are not counts of independent biological discoveries. Source-quality tiers remain attached to the estimates. Completing the calculations did not clear excluded sources or promote sensitivity-tier results to primary evidence.

Reproduction from historical processed inputs is complete, while verification from original source bodies through harmonization and preprocessing is incomplete. At drafting, ordered scientific content and QC comparisons for all 13 validation sources had completed with independent qualified agreement. Compressed bytes differed, while the ordered scientific content and QC matched; the historical nine necessary-QC passing and four failing source classifications were retained. Two separate core comparison attempts completed 5 of 27 and 7 of 35 source chains, respectively, and the extension attempt completed 79 of 100; each stopped before full closure. These counts describe separate plans and are not combined. Thus, agreement of the native calculations presently supports numerical reproduction conditional on the admitted processed inputs, with validation-source comparisons providing additional provenance evidence. The available raw-source evidence remains partial and does not establish complete original-source reconstruction.

Sharing is substantial and phenotype specific

Within the core panel, insomnia shared common-variant effects with major depressive disorder (genetic correlation, $r_g = 0.442$; marginal 95% confidence interval [CI], 0.393–0.492), frailty (0.641; 0.594–0.687) and body mass index (BMI; 0.172; 0.127–0.217). Coded sleep apnoea showed a larger BMI correlation point estimate (0.524; 0.482–0.565). These are primary-tier FDR positives in the complete core family (Fig. 2, Supplementary Figs. 1–2 and Supplementary Data 1). The joint pattern is compatible with broad sharing across mood, general health and adiposity; it does not distinguish pleiotropy, behavioural pathways, ascertainment or causal direction.

Gastrointestinal labels illustrated the limits of a broad disease grouping. In the extension, insomnia correlated positively with abdominal pain (0.559; 0.489–0.629), gastro-oesophageal reflux (0.522; 0.447–0.596) and noninfectious gastroenteritis (0.521; 0.424–0.619), each passing the frozen extension FDR threshold. In contrast, its core inflammatory bowel disease correlation was negative (–0.074; –0.133 to –0.015; core FDR $q = 0.0364$). These source-specific findings do not support a uniform inflammatory-gut interpretation. Symptoms, diagnostic categories and immune disease require separate clinical interpretation, and correlated coding or healthcare contact could contribute to the observed pattern.

For reflux, continuous reported duration had a negative correlation (–0.193; –0.269 to –0.117), whereas short sleep relative to 7–8-hour controls and long sleep relative to the same control category had positive correlations (0.382; 0.300–0.464 and 0.197; 0.100–0.295, respectively). All three passed the extension FDR threshold. These categorical contrasts do not estimate a continuous dose–response curve and cannot specify an optimal sleep duration or a clinical prescription. Shared controls and overlapping participants also prevent treating the tail results as independent confirmations.

Device duration and reflux had an estimate near zero with an interval spanning both directions (–0.002; –0.091 to –0.087), while device efficiency was negatively correlated (–0.135; –0.229 to –0.040; extension FDR positive; Fig. 3). These are descriptive observations about distinct source-defined traits. A significant questionnaire association and a nonsignificant device association do not establish a difference between them. Their shared disease source and overlapping sleep cohorts require joint sampling covariance for such a test. No calibrated questionnaire–device contrast is reported here.

External evidence replaces the outcome source

The historical external family contained 217 candidate pairs. Forty-one genetic correlations were available: 23 qualified external outcome-side positives met the frozen criterion, and 18 were directionally concordant but below its threshold. Seventeen other candidates were QC-ineligible, and 159 lacked eligible available source evidence. The 17 QC exclusions comprised nine failures of the heritability signal criterion and eight intercept exclusions. These categories preserve missingness and source limitations rather than assigning numerical nulls to unestimated pairs (Fig. 4 and Supplementary Data 3).

The 23 qualified positives supported selected patterns across alternate outcome sources. For example, insomnia correlated with MVP abdominal pain at 0.410 (0.345–0.475), FinnGen R13 reflux at 0.364 (0.290–0.438), diaphragmatic hernia at 0.192 (0.115–0.268), and other noninfectious enteritis or colitis at 0.306 (0.188–0.424). These findings passed the original external criterion, including direction concordance and $P < 0.05/217$. Their point estimates cannot be compared formally with the discovery estimates using an independent-estimates formula.

All 41 estimates reused the original sleep GWAS. Consequently, zero pairs established fully independent replication with both a new sleep source and a new outcome source. Distinct cohort names also do not prove

absence of shared participants or relatives. Cohort intersections, reuse across releases and compatibility of outcome definitions remain relevant. The available validation evidence supports bounded outcome-source robustness; it does not validate a clinical sleep diagnosis, isolate measurement effects or establish transport to other ancestry groups. Prior-art review classified 15 of the 23 positives as substantially similar to earlier findings, six as related, and two as unresolved. Unresolved priority does not establish novelty.

Sensitivities clarify regression and scale dependence

Sixty-two additional sensitivity estimates were completed and independently checked: 36 lipid genetic correlations under an explicit two-step threshold of 30, 24 binary-outcome sample-size correlation estimates and two standalone heritability estimates. The lipid analysis preserved sources and SNP universes. Both pair-specific heritability intercepts were at or below 1.2 in none of the 36 baseline fits and in 35 of the 36 sensitivity fits; all 36 retained both heritability Z statistics at or above four. The sleep-apnoea–HDL fit retained an outcome intercept of 1.2067, above the historical boundary. Improved intercept diagnostics did not independently establish source validity or change the frozen primary classification.

The largest absolute lipid-correlation change was 0.0292. Device-duration–HDL moved from -0.0497 (-0.0954 to -0.0040) to -0.0284 (-0.0643 – 0.0074), so its sensitivity interval included zero (Fig. 5). Insomnia–HDL remained negative under the sensitivity specification (-0.1175 ; -0.1494 to -0.0857). These estimates quantify specification sensitivity, without a calibrated between-fit difference test or a new FDR screen.

For multiple sclerosis and melanoma, effective-sample-size and total-sample-size presentations changed heritability scale while genetic-correlation point estimates changed by at most 5.1×10^{-15} . Melanoma's heritability Z statistic remained approximately 3.568 and therefore below the frozen threshold. Its exclusion was not rescued by a numerically plausible rescaling. Some historical liability presentations exceeded one; alternative presentation conventions brought values into a physical range, but this arithmetic did not establish a valid population prevalence or corrected population heritability. Sample size, case fraction and prevalence must describe the same estimand and ascertainment scheme.

Clinical insomnia feasibility remains qualified

A separate FinnGen R13 F5_INSOMNIA analysis used 51,643 cases and 446,273 controls, with assumed constant effective $N = 185,146.704$. Its observed-scale reporting estimate was $h^2 = 0.05354$ (SE 0.00407; marginal 95% CI, 0.04556–0.06151), with heritability $Z = 13.152$ and intercept 1.0779. This established sufficient measured signal for the completed feasibility analysis under its assumptions. It did not produce a new sleep–disease correlation, certify population liability-scale heritability or independently validate the historical atlas.

The source supplied an explicit log-odds effect and standard error, but INFO and per-variant N were unavailable. Finnish reference suitability, sample overlap and clinical transport remain qualified. Its coded insomnia phenotype and sleep-disorder exclusions differ from frequent questionnaire complaints. The MVP clinical-insomnia candidate also requires resolution of effect/CI-derived standard-error semantics and clinical ascertainment details. Neither candidate currently supplies an admitted independent replacement for the questionnaire exposure.

Discussion

The reconstructed atlas provides a numerically reproduced account of common-variant sharing across observed sleep constructs. Its strongest supported biological description is heterogeneous sharing across mood, frailty, adiposity and selected gastrointestinal phenotypes. The opposing inflammatory bowel disease and

symptom-related directions show why a pooled gastrointestinal label can obscure rather than explain the evidence. Likewise, duration tails and device-derived measurements describe distinct phenotypes; their differing marginal results should guide study design rather than be converted into clinical equivalence, optimal-duration or modality-specific biological claims.

These observations sit within an established literature on sleep domains and health sharing [5–7]. The present work does not establish a newly discovered association or a substantive new statistical solution. Its contribution is a traceable reconstruction that separates numerical agreement, diagnostic robustness, outcome-source validation and currently unsupported interpretation. Whether that contribution meets a particular journal's threshold for scientific advance requires independent human assessment.

Several limitations are consequential. First, the historical screens and validation set were already observed. Illustrative associations are selected for explanation after inspection, so their magnitude cannot be treated as an unbiased estimate from a newly selected discovery programme. Winner's curse and correlated phenotype selection remain possible. The full families, excluded tiers and unavailable candidates are retained to make that selection visible. Additional candidate sources or alternative outcomes cannot be regarded as unlimited attempts under the original validation denominator.

Second, a genetic correlation is a normalized genetic covariance. Classical independent measurement noise need not mechanically attenuate that correlation, although it can reduce heritability and precision. Dividing an observed genetic correlation by assumed phenotype reliability would therefore be unjustified. Genetically structured reporting or diagnostic error can alter genetic moments, but the current summary statistics do not identify its decomposition from biological liability. Questionnaire recall, inclusion of naps, device inference, elapsed measurement time and participation differ together. A pure modality or measurement-error effect would require anchored or crossed measurements and an identified model.

Third, uncertainty between related estimates remains unresolved. Established multivariate approaches propagate joint sampling covariance [8]. Separately numbered block-jackknife outputs do not establish common genomic deletions across fits, and marginal standard errors alone do not justify a correlation-difference test. The proposed finite questionnaire/device-duration comparison with FinnGen reflux and chronic obstructive pulmonary disease was not performed: the existing validation table contains no device-duration estimates, so even its complete marginal standard-error set is absent. The inspected multiorgan study already reports parallel COPD correlations for named Jones 2019 device-duration and Dashti 2019 questionnaire-duration sources [12]. Those marginal precedents constrain novelty without providing a verified dependent contrast. No calibrated heterogeneity, causal or latent-construct inference follows from the current atlas.

Fourth, outcome-side evidence does not remove sleep-source selection or establish ancestry transport. UK Biobank participation can distort genetic analyses [9], and device participation introduces a further selected subset. FinnGen and MVP broaden outcome ascertainment but differ in population, sex composition, care setting and coding [10,11]. Broad European-ancestry labels do not certify a suitable LD reference for every source or generalizability to other populations. Clinical source contracts require exact event rules, control exclusions, observation periods and case-count semantics. An underpowered or QC-ineligible phenotype cannot serve as evidence of biological absence.

A future study should state its clinically meaningful target and minimum informative precision before examining new results, document the relevant cohort intersections and source-specific effective sample sizes, and secure validation that matches its intended claim. A bounded estimate near zero can support limited robustness only when its interval excludes a prespecified meaningful difference. Conversely, a calibrated difference between source-defined correlations would still require competing explanations and compatible

replication. These requirements are unmet here, and the unfinished raw-source comparisons must be resolved before the reproduction package is finalized.

Methods

Source defined panel and reconstruction

The historical panel and accession-specific metadata were retained without replacing sources with newer releases. Core analyses paired 12 sleep traits with 33 outcomes; extension analyses used the same 12 traits against 100 phenotypes. Supplementary Data 5 provides the literal 12-source construct dictionary, including releases, observed definitions, selected effect/uncertainty columns, directions, sample counts, control rules and unresolved fields. The public insomnia input was the UK Biobank-only Jansen release, not the complete UK Biobank/23andMe meta-analysis [1]. Continuous reported duration concerned whole-hour 24-hour sleep including naps. Short sleep was below seven hours versus seven to eight hours; long sleep was at least nine hours versus the same control category [2]. Morningness was a morning-person versus evening-person contrast. Habitual snoring asked whether a partner, close relative or friend complained about the participant's snoring. Sleep apnoea used the original FinnGen R9 hospital/death-registry endpoint and control exclusions; its code/control qualifications, including an unresolved historical ICD-9 anomaly, preclude an unqualified interpretation as pure obstructive apnoea. Device efficiency, duration and midpoint retained their selected rank-normalized Jones 2019 transforms. Device duration concerned inferred sleep within the main sleep-period window, measured over up to seven wear days in a selected subset 2.8–9.7 years after baseline [3]. Quiet wake and sleep outside that window are relevant measurement limitations. These operational definitions were not equated with diagnosed insomnia, polysomnographic sleep or each other.

Source records retain ancestry, build, effect/allele semantics, source N, case/control counts, SNP selection and reference dependencies. Historical source-specific filtering, harmonization and munging rules were preserved, rather than asserting an identical INFO filter across sources. One original core source container, for prostate cancer, remains unrecovered despite an available verified historical materialized raw-stage input. Original-body recovery and processed-input admission are distinct from the ordered preprocessing comparisons. No new association fits were undertaken for manuscript preparation. Preserved partial core attempts (5 of 27 and 7 of 35 chains) and the extension attempt (79 of 100) leave full source-chain verification incomplete; these separate plans are not pooled.

Genetic correlation and uncertainty

We used the recorded native LDSC implementation at CBIIT revision 6c673952cee74bd5c57aef1555a03b1c015399a0, with Python 3.9.23, NumPy 1.21.5, pandas 1.3.3 and SciPy 1.7.3. Reference inputs and commands were bound to the reproduction records. Genetic correlation divides the common-variant genetic covariance by the square root of the product of the two common-variant heritabilities, describing standardized sharing for the analysed phenotypes under LDSC assumptions [4]. Historical unconstrained bivariate settings were retained; the lipid sensitivity explicitly changed the two-step threshold to 30. We retained the recorded within-fit block-jackknife standard errors and calculated marginal normal 95% intervals as estimate ± 1.96 SE. These intervals are not simultaneous family-wide intervals and are not confidence intervals for differences between fits.

Native full-precision capture was checked against native summaries, available historical printed logs and independent arithmetic reviews. Historical unrounded quantities were unavailable. Numerical agreement therefore refers to the observable historical precision. Agreement under a processed-input reconstruction does not prove every original-body preprocessing step, which remains partly pending.

Eligibility and multiplicity

Historical source eligibility used standalone heritability $Z \geq 4$ and intercept ≤ 1.2 . Heritability components and intercepts estimated within bivariate fits were tracked separately as pair-specific sensitivity diagnostics. Original tiers and exclusion reasons were retained; passing a pair-specific diagnostic or a later sensitivity did not overwrite the original standalone source classification. Benjamini–Hochberg correction was retained separately across all 396 core and all 1,200 extension tests. The primary and sensitivity-tier counts are partitions of the core family's positives, not separately recomputed significance families. No sensitivity run created an additional discovery family.

For the 217 historical external candidates, a qualified positive required the original source/QC criteria, direction concordance and the frozen $P < 0.05/217$ threshold. Directional results below that threshold, QC exclusions and unavailable source evidence were retained as distinct categories. All estimated pairs reused the original sleep source; none was labelled fully independent two-trait replication. The reported 95% intervals are marginal even when a point estimate passes the more stringent external threshold.

Sample size scale and transport assessment

For a single case–control cohort, effective sample size was derived by multiplying four times the case count by the control count and dividing by their combined total. Effective-N input, total-N input, sample case fraction and population prevalence are different quantities. Liability-scale presentation requires a compatible ascertainment convention. Its positive multiplier divides the squared population prevalence times the squared population noncase fraction by the product of the sample case and noncase fractions and the squared standard-normal density at the liability threshold. That threshold has an upper-tail probability equal to the specified population prevalence. The same multiplier scales heritability and its standard error. For meta-analyses, aggregate counts do not automatically reproduce cohort-specific information. Historical prevalence conventions were retained for the diagnostic comparisons and were not newly certified as population values.

Clinical feasibility and external interpretation were assessed from accession-specific definitions, effect and SE documentation, INFO/N availability, ancestry/reference suitability and possible within- and cross-stage cohort intersections. A clinical code match was not assumed to establish equal chronicity, diagnostic sensitivity or control selection. Prior publications and their inspected supplements were used to qualify novelty; inaccessible or uninspected evidence was retained as unresolved rather than proof that an association was unreported. Automated specialist reviews examined statistical, clinical, biological and identification limits; they are not substitutes for accountable human scientific approval.

Data and code availability human completion required

[Investigators must specify the permitted repository/DOI, accession-specific access routes, source attribution and restrictions, and the final source-chain status. Aggregate evidence, scripts and figure source mappings exist in the local research package; public deposition and redistribution rights have not been established. Do not substitute a planned archive for an available deposited resource.]

Human declarations unresolved placeholders not findings

- **Authors, affiliations, corresponding author and contributions:** [Human investigators must confirm authorship eligibility, affiliations, order, correspondence and accountable contributions.]

- **Ethics and consent:** [Human investigators must determine the institutional requirements for this secondary analysis and provide source-specific original approvals/consent and any required current approval or exemption. No new approval or exemption is asserted.]
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- **AI use disclosure for human verification:** Codex and automated reviewer agents assisted source recovery and provenance work, code generation and correction, statistical evidence checking, plotting, literature/evidence review, and preparation of this draft. The inspected record identifies the GPT-6 model family; exact model/version histories require human confirmation. Automated checks and specialist-style reviews occurred, but accountable human validation and approval are not documented in the inspected record. Investigators must verify all scientific content, references, analyses and declarations and finalize the disclosure; AI systems are not authors.

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Main figure legends

Figure 1 | Scope of the reproduced evidence. The evidence flow summarizes 190 native jobs, 158 standalone heritability estimates and 1,637 genetic correlations, including the frozen core, extension and historical external families. Its 23 qualified and 18 directional external results reuse the sleep GWAS. The 17 exclusions labelled “QC/power” comprise nine heritability-Z and eight intercept exclusions; they are not all failed power calculations. Unavailable pairs are not null tests. This figure documents completed native numerical calculations, not complete raw-source preprocessing or independent replication. Source data combine the full-precision stage tables, the 217-row classification and completion records.

Figure 2 | Core genetic sharing and marginal precision. Heatmaps display 132 of the complete 396 core sleep–outcome pairs, including major depressive disorder and inflammatory bowel disease; the remaining core outcomes, including frailty and BMI, are in Supplementary Figs. 1–2. Panel A shows genetic correlation and panel B its standard error. Dots denote $FDR < 0.05$ in the frozen complete 396-test family; original sensitivity-tier markings remain as supplied. Colour and marginal significance do not test differences between sleep traits or identify causal mechanisms. Full values, intervals, historical tiers and source diagnostics appear in Supplementary Data 1.

Figure 3 | Extension patterns include distinct reflux constructs. Heatmaps display 156 pairs from the complete extension family, including reflux; the remaining outcomes are in Supplementary Figs. 3–9. Panel A shows genetic correlation and panel B its standard error. Dots refer to $FDR < 0.05$ in the entire 1,200-test family. Questionnaire duration, duration categories, coded sleep apnoea and device measurements retain their source-defined meanings. A nonsignificant device-duration estimate does not establish absence of sharing or a difference from questionnaire duration. Supplementary Data 2 contains all 1,200 results.

Figure 4 | Historical external outcome-source estimates. The interval plot shows 14 of the 41 available estimates, with point estimates and marginal normal 95% intervals; remaining estimates are in Supplementary Figs. 10–11. The complete family contains 217 candidates, including QC-ineligible and unavailable pairs. Crossing or excluding zero in this display does not determine qualification under $P < 0.05/217$. Classification also retains source/QC and direction criteria. All displayed fits reuse the historical sleep GWAS; none is a fully independent two-trait replication. Supplementary Data 3 supplies estimates and the complete classification.

Figure 5 | Lipid sensitivity preserves uncertainty and source qualifications. Paired interval plots display 12 baseline and explicit two-step-30 sensitivity correlations involving coded sleep apnoea and the three device traits. Points and bars are estimates and marginal normal 95% intervals. Other lipid pairs are Supplementary Figs. 12–13. No between-fit P value or new FDR family is supplied. The device-duration–HDL sensitivity interval includes zero, and the sleep-apnoea–HDL outcome intercept remains above 1.2. Supplementary Data 4 binds all 62 sensitivities and the liability presentations.

Supplementary display and source data plan outside the manuscript body

The existing 23-page figure set is retained without new result selection or numerical rendering. Main Figs. 1–5 use index identifiers 21, 1, 4, 12 and 17, respectively, from the `selected_page` field in `scientific_figure_index_v4_4.tsv`; these are not page numbers in a combined PDF. Supplementary Figs. 1–18 contain the other indexed pages in the mapping file: the remaining core/extension/validation/lipid pages; two binary sample-size pages; the liability-presentation page; and the two explicitly uncalibrated covariance-zero diagnostic pages. The last two are conditional diagnostic illustrations and cannot support calibrated heterogeneity claims. The figure mapping binds each existing PDF/SVG/PNG and its hash to numerical tables and explains the evidence-flow figure's multiple count sources. Supplementary Data 1–4 are proposed labels for the existing complete aggregate evidence; Supplementary Data 5 is the 12-source construct dictionary. These labels do not assert that a journal repository has already been deposited.